

**Cairo University**  
**Faculty of Engineering**  
**Department of Computer Engineering**



# **Baligh**

**Arabic Writing Assistant**

A Graduation Project Report Submitted

to

Faculty of Engineering, Cairo University

in Partial Fulfillment of the requirements of the degree

of

Bachelor of Science in Computer Engineering.

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# Abstract

Arabic digital writing still suffers from a shortage of integrated tools that can correct Modern Standard Arabic text accurately, respond in real time, and explain their feedback in a way that helps users learn. Existing tools often focus on surface spelling fixes or black-box suggestions without exposing the grammatical reasoning behind the correction. Baligh addresses this problem by providing an Arabic writing assistant that combines grammatical error detection, grammatical error correction, spelling-aware editing, next-word suggestion, and explanatory feedback in a single user-facing system. The project aims to support students, educators, and Arabic content writers by improving writing quality while also making the correction process more understandable and educational.

The implemented system follows a modular pipeline. A preprocessing layer performs normalization, word-boundary handling, segmentation, and morphological analysis and disambiguation. On top of that, Baligh integrates a grammatical error detection engine that fuses rule-based, lexicon-based, and machine-learning detectors, then passes its findings to correction components built around ontology-guided reasoning, dictionary lookup, and text-editing models. In parallel, a next-word suggestion module supports both next-word prediction and word auto-completion. The final system highlights issues, proposes corrections, returns ranked suggestions, and presents grammar-oriented explanations to the user in an interactive workflow.

The project outputs include the preprocessing pipeline, grammatical error detection and correction services, the next-word suggestion subsystem, and a working user application. Testing and validation relied on automated unit and integration tests together with evaluation on established Arabic linguistic resources, including QALB14, QALB15, and ZAEBUC. Overall, Baligh demonstrates that a modular and explainable Arabic writing assistant can combine linguistic processing and machine learning into a practical platform for improving Arabic writing quality.

## الملخص

لا تزال الكتابة الرقمية باللغة العربية تعاني من نقص الأدوات المتكاملة التي تستطيع تصحيح نصوص العربية الفصحى بدقة، والعمل بزمان استجابة مناسب، وتقديم تفسير واضح يساعد المستخدم على التعلم بدل الاكتفاء بإظهار التصحيح فقط. فكثير من الأدوات المتاحة تركز على الأخطاء الإملائية السطحية أو تقدم اقتراحات غير مفسرة. يعالج مشروع بليغ هذه المشكلة من خلال بناء مساعد كتابة عربي يجمع بين اكتشاف الأخطاء النحوية، وتصحيحها، ومعالجة الأخطاء الإملائية، واقتراح الكلمة التالية، وتقديم تفسير مبسط للقاعدة أو سبب الخطأ داخل نظام واحد موجه للمستخدم. ويهدف المشروع إلى خدمة الطلاب والمعلمين وصناع المحتوى العربي عبر تحسين جودة الكتابة وجعل عملية التصحيح أكثر فائدة من الناحية التعليمية.

يعتمد النظام المنفذ على بنية معيارية تبدأ بطبقة للمعالجة المسبقة تشمل التطبيع، وتحديد حدود الكلمات، والتجزئة، ثم التحليل الصرفي وفك الالتباس. بعد ذلك يدمج بليغ محركاً لاكتشاف الأخطاء النحوية يعتمد على المزج بين قواعد لغوية، وكشافات معجمية، ونماذج تعلم آلي، ثم يمرر النتائج إلى وحدات التصحيح المبنية على الاستدلال المعتمد على الأنطولوجيا، والبحث المعجمي، ونماذج التحرير النصي. وبالتوازي مع ذلك، يوفر النظام وحدة لاقتراح الكلمة التالية تشمل التنبؤ بالكلمة التالية والإكمال التلقائي للكلمة الجارية كتابتها. وتعرض المنصة النهائية الأخطاء والتصحيحات والاقتراحات والتفسيرات اللغوية بصورة تفاعلية.

تشمل مخرجات المشروع خط معالجة مسبقة متكامل، وخدمات لاكتشاف الأخطاء وتصحيحها، ووحدة اقتراح الكلمات، وتطبيقاً عملياً للمستخدم. واعتمد التحقق من النظام على اختبارات آلية للوحدات والتكامل، إضافة إلى تقييم مبني على موارد عربية معروفة مثل QALB14 و QALB15 و ZAEUC. وتُظهر المحصلة أن بليغ يقدم نموذجاً عملياً لمساعد كتابة عربي قابل للتفسير يجمع بين المعالجة اللغوية والتعلم الآلي لخدمة الكتابة العربية بصورة أكثر جودة وفائدة.

# Acknowledgment

*Write your acknowledgment and dedication here.*

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# List of Abbreviations

*Put abbreviations in alphabetical order.*

| Abbreviation | Meaning                      |
|--------------|------------------------------|
| AI           | Artificial Intelligence      |
| GEC          | Grammatical Error Correction |
| GED          | Grammatical Error Detection  |
| MSA          | Modern Standard Arabic       |
| NLP          | Natural Language Processing  |
| NWP          | Next-Word Prediction         |

# List of Symbols

*Put symbols in alphabetical order. Greek symbols should come first, followed by English symbols.*

| Symbol   | Meaning                                |
|----------|----------------------------------------|
| $\alpha$ | Learning rate or weighting coefficient |
| $P$      | Probability                            |
| $T$      | Time or sequence length                |

# Contacts

## Team Members

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# Chapter 1: Introduction

This chapter introduces the project, justifies the need for it, states the project outcomes, and explains the organization of the report.

## 1.1. Motivation and Justification

*Explain why the project is needed, what problem it addresses, why the problem is significant, and who benefits from the project.*

Arabic writing tools often provide corrections without explaining the grammatical reasoning behind them. Baligh focuses not only on correction, but also on understandable explanations that help users improve their writing over time.

## 1.2. Project Objectives and Problem Definition

*State the project objectives clearly and end with a formal problem definition.*

## 1.3. Project Outcomes

*State the final project outcomes: models, datasets, applications, APIs, experiments, reports, or other deliverables.*

## 1.4. Document Organization

*Briefly describe the following chapters.*

## Chapter 2: Market Visibility Study

This chapter surveys the market related to the project and discusses similar tools or platforms. It should explain their limitations and justify the need for Baligh.

### 2.1. Targeted Customers

*Mention the intended customers/users and how they benefit from the project.*

### 2.2. Market Survey

*List competing or related products. Discuss each one in a subsection and explain its pros and cons.*

#### 2.2.1. Competitive Product 1

*Describe the product, strengths, weaknesses, and relevance to Baligh.*

#### 2.2.2. Competitive Product 2

*Describe the product, strengths, weaknesses, and relevance to Baligh.*

### 2.3. Business Case and Financial Analysis

*Describe the possible business case, expected users, pricing assumptions, CapEx, OpEx, revenue, cash flow, and break-even point.*

Table 2.1: Sample Five-Year Financial Projection

| Year | Expected Users | Revenue | Expenses | Profit Before Tax |
|------|----------------|---------|----------|-------------------|
| 1    | —              | —       | —        | —                 |
| 2    | —              | —       | —        | —                 |
| 3    | —              | —       | —        | —                 |
| 4    | —              | —       | —        | —                 |
| 5    | —              | —       | —        | —                 |

## Chapter 3: Literature Survey

This chapter should include necessary engineering and non-engineering background, followed by a concise literature review of recent work related to the project.

### 3.1. Background on Arabic NLP

*Explain the most important concepts needed to understand the project, such as morphology, spelling correction, grammatical error correction, and language modeling.*

### 3.2. Background on Writing Assistance Systems

*Explain writing assistants, correction pipelines, feedback generation, and user-facing explanations.*

### 3.3. Comparative Study of Previous Work

*Classify and compare previous work. Focus on relevance, strengths, limitations, datasets, methods, and results.*

**Table 3.1:** Comparison of Related Arabic Writing Assistance Approaches

| Work/System | Main Approach | Strengths       | Limitations                 |
|-------------|---------------|-----------------|-----------------------------|
| System 1    | Rule-based    | Explainable     | Limited coverage            |
| System 2    | Neural        | Strong accuracy | Less interpretable          |
| System 3    | Hybrid        | Balanced        | Requires integration effort |

### 3.4. Implemented Approach

*State the approach chosen from the reviewed approaches and justify why it was selected. Mention any modifications added by the team.*

# Chapter 4: System Design and Architecture

This chapter is the main body of the report. It should answer: what has been done, and how has it been done? The discussion should flow from the overall architecture to modules and submodules.

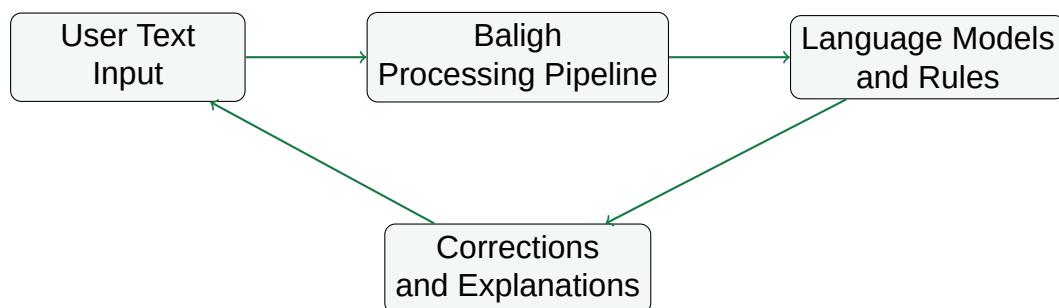
## 4.1. Overview and Assumptions

*Introduce the system design and list the assumptions used during design and implementation.*

## 4.2. System Architecture

*Describe the overall architecture and how modules communicate.*

### 4.2.1. Block Diagram



**Figure 4.1:** High-level architecture of the Baligh system.

## 4.3. Module 1: Grammatical Error Detection

### 4.3.1. Functional Description

*Explain what this module does.*



#### **4.3.2. Modular Decomposition**

*Break the module into submodules.*

#### **4.3.3. Design Constraints**

*Explain constraints such as latency, Arabic ambiguity, model size, and data availability.*

#### **4.3.4. Other Description of Module 1**

*Add any other relevant technical details.*

### **4.4. Module 2: Correction and Explanation Generation**

#### **4.4.1. Functional Description**

*Explain correction generation and explanation generation.*

#### **4.4.2. Modular Decomposition**

*Break the module into submodules.*

#### **4.4.3. Design Constraints**

*Explain constraints such as explanation quality, grammatical accuracy, and user readability.*

#### **4.4.4. Other Description of Module 2**

*Add any other relevant technical details.*

# Chapter 5: System Testing and Verification

This chapter describes how the system was tested and verified.

## 5.1. Testing Setup

*Describe hardware, software, datasets, versions, and test environment.*

## 5.2. Testing Plan and Strategy

*Describe unit tests, integration tests, user testing, model evaluation, and manual review.*

### 5.2.1. Module Testing

*Explain how each module was tested independently.*

### 5.2.2. Integration Testing

*Explain how the full pipeline was tested after integration.*

## 5.3. Test Schedule

*Add the timeline/schedule of testing activities.*

## 5.4. Comparative Results to Previous Work

*Compare your results with baselines or previous work where applicable.*

# Chapter 6: Conclusions and Future Work

## 6.1. Faced Challenges

*Discuss technical, management, research, data, and integration challenges.*

## 6.2. Gained Experience

*Discuss technical and non-technical learning outcomes.*

## 6.3. Conclusions

*Summarize what the project achieved and why it matters.*

## 6.4. Future Work

*Mention possible future improvements and extensions.*

# References

[1] Author Name, "Paper or Book Title," *Conference or Journal Name*, Year.

[2] Author Name, "Another Reference Title," Publisher or Venue, Year.

# Chapter A: Development Platforms and Tools

*List programming languages, frameworks, libraries, operating systems, hardware, datasets, and development tools.*

## Chapter B: Use Cases

*Describe system use cases, actors, preconditions, main flows, alternative flows, and expected results.*

## Chapter C: User Guide

*This appendix should be included. Explain how to install, run, and use the final system. Add screenshots or step-by-step instructions when available.*

## Chapter D: Code Documentation

*Document important modules, classes, APIs, configuration files, and code structure.*



## Chapter E: Feasibility Study

*This appendix should be included. Add market, technical, operational, and financial feasibility details.*